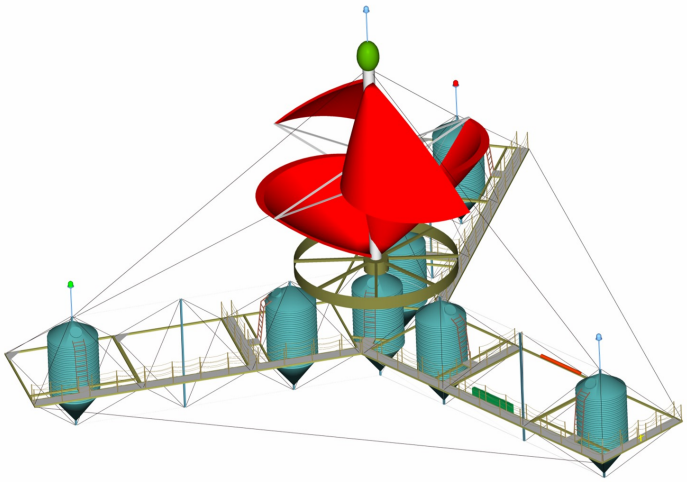
.11/4/2004

B.E.A.R. Oceanics Llc.

GENERAL SPECIFICATION

OCEAN GYPSY

ETHANOL PRODUCING PLATFORM



NDI ENGINEERING COMPANY
Research and Consulting Engineers
100 Grove Road, P.O. Box 518 ● Thorofare, New Jersey 08086
(856) 848-0033 ● (215) 564-6145 ● FAX (856) 848-0277 ● www.ndieng.com

FOREWORD....5

G01 GENERAL DESCRIPTION OF THE PLATFORM....6

G02 MAIN PARTICULARS....6

1 GENERAL PRINCIPLES....8

10 Rules and Regulations....8

11 Tests....8

2 PLATFORM STRUCTURE....9

20 Main Support Structure....9

21 Tension Cables....9

22 Flotation Tanks....9

23 Vertical Support Members....9

3 PROCESS EQUIPMENT AND SYSTEM....11

30 Wind Turbine....11

31 Bull Gear....12

32 Electrolyser....12

33 Process Gas Compressor....13

34 Electric Generator....13

35 Reactor....13

36 Condenser....13

39 Process Control System....14

38 Bladder Storage Tank....14

4 PLATFORM EQUIPMENT....15

41 Cleats....15

42 Mooring Bitt....15

43 Bladder Tank Towing Bitts....15

44 Electrical System....15

45 Storage Batteries....15

46 Navigational Lighting....15

5 MAJOR EQUIPMENT LIST....17

51 Platform Structure....17

52 Process Equipment....17

53 Platform Equipment....18

FOREWORD

When used in conjunction with the documents listed below, this specification describes a floating ethanol producing platform and its systems to the level of detail sufficient to understand its main functions and characteristics.

Reference Documents

- General Arrangement Drawing No. 1157-001-01
- Ethanol Process Flow Diagram No. 1157-001-02
- Ethanol Piping Diagram No. 1157-001-03
- Weight and Moment Study

This platform is a prototype and, when constructed, will be used to ascertain the soundness of the design and functionality of the installed process system and its components.

G01 GENERAL DESCRIPTION OF THE PLATFORM

The platform is a free-floating vessel, un-manned and entirely self-contained, the purpose of which is to use wind energy to convert seawater and carbon dioxide into ethanol.

The platform is configured in a "Y" arrangement, with flotation tanks installed at either end of each leg and a center tank in the center section. In addition to providing flotation, the three inboard leg tanks and center tank support a wind turbine and house the rotating machinery used to produce the ethanol and provide electrical power.

The platform is designed to be towed from the end of one of its three legs – the leg opposite the storage bladder. Consequently, there is a forward, port, and starboard leg on the "Y" platform.

A structural mast, extending from the center tank, serves a dual purpose of housing the system's electrolyser, in which seawater will be separated into its constituent elements, hydrogen and oxygen, and providing a support for the upper bearing of the wind turbine.

The platform itself is a tensegrity structure utilizing four inch square tube and ¾ inch fiber core wire rope to provide an extremely strong structure with relatively light weight. The flotation tanks are commercially available grain silos wrapped in a fiberglass/epoxy coating that provides for a watertight boundary capable of withstanding high impact loads.

The wind turbine is a vertical axis type with buckets lined with aluminum to ensure good radar visibility. The turbine, with a 30 foot bull gear attached to the lower end of the center shaft, rotates about the center mast at a speed of 23 RPM. The bull gear, in turn, engages sprockets which turn vertical input shafts on an electric generator and a process gas compressor, each of which is located in one of the inboard leg tanks surrounding the central tank.

Ethanol, the final product, is stored in a floating 40,000 gallon bladder tank, attached to the platform by wire rope. A flexible hose connecting the platform process piping to the bladder tank is used to fill the tank by residual pressure at the last step in the production process.

It is intended that this prototype platform be anchored in place during production with the bladder tank periodically being emptied.

G02 MAIN PARTICULARS

Length overall (w/o bladder tank)	120 feet

Length overall with bladder tank (approx.)	180 feet
Height above water	79 feet
Draft	6 feet
Displacement	6 tons
Storage capacity	40,000 gallons
Product throughput (maximum)	40,000 gpd

1 GENERAL PRINCIPLES

The platform is designed such that it can be assembled and disassembled with a minimum of lifting equipment, which shall consist primarily of mobile cranes and forklifts.

10 Rules and Regulations

The platform and its installed machinery is to be designed and constructed in such a manner as to receive a United States Coast Guard Certificate of Inspection.

11 Tests

All installed piping will be hydrostatically tested in accordance with 46 CFR 56.97.

The following shall be demonstrated to be operable as designed to the satisfaction of the local USCG inspector:

Machinery and Process Equipment

Controls and Safety Systems

Mooring and Towing Equipment

Navigation Lighting

Communications, Telemetry, and Monitoring Systems

2 PLATFORM STRUCTURE

20 Main Support Structure

The main support structure is fabricated and assembled into a "Y" shaped configuration. Structural members are made from 4 inch square tubing, and will be coated inside and out with a corrosion preventing coating system.

All joints in the structural frame work are welded with the exception of the center section where the three legs meet. These joints are bolted to facilitate assembly, disassembly, and transportation

21 Tension Cables

Tension cables connect the flotation tanks and vertical support members (section 23) to the main support structure. Each cable is provided with a turn buckle to ensure consistency in setting tension.

All cables are to be coated with a corrosion protection coating system after assembly.

22 Flotation Tanks

Flotation tanks are commercially available grain silos, manufactured from 20 gage corrugated steel that has been galvanized.

An access hatch will be provided on the upper surface of each tank, accessible by a ladder from the main structural support frame.

After assembly, each flotation tank will be wrapped with a fiberglass and epoxy coating system to ensure water tight integrity and protection against impact from floating debris.

The inside of each tank will be coated with a corrosion preventing coating system.

23 Vertical Support Members

Vertical support members are fabricated from 4 inch square tube with eye-bolts welded to each surface at both ends the purpose of which is to provide anchoring points for the tension cables.

Vertical support members installed through the flotation tanks are to be fitted through a welded flange plate and bolted to the tank flanges top and bottom with a water tight gasket.

After fabrication, vertical support members will be coated inside and out with a corrosion preventing coating system.

3 PROCESS EQUIPMENT AND SYSTEM

The ethanol production process is designed to be completely self contained and automated, requiring no manual control or input with the exception of periodically emptying the storage tank and replacing a carbon dioxide (CO_2) cylinder.

All energy required by the processing system will be derived from a wind powered turbine which will be directly connected to a gas process compressor and an electric generator.

The inlet to the compressor is connected to the center mast which, as a result, will be placed under a negative pressure. The center mast which is open and submerged at the bottom contains seawater and positive and negative direct current electrodes. The combined effect of the low pressure and electric current separates the seawater into hydrogen and oxygen which then is pulled into the process gas compressor and sent through the piping system. CO_2, supplied from a high pressure storage tank, is injected into the hydrogen and oxygen stream before it enters a reactor.

Inside the reactor, the hydrogen, oxygen, and carbon radicals combines to form gaseous ethanol (grain alcohol). The gaseous ethanol will then be sent to an air cooled condenser permitting the ethanol to liquefy.

The pressure provided at the discharge of the compressor will be sufficient to move the ethanol through the process system and into a bladder type storage tank, tethered to the stern of the platform.

The throughput of ethanol is directly related to the energy produced by the wind turbine; i.e. low wind speeds will produce a lower ethanol output than higher wind speeds. In the event the system pressure drops to a low enough point to where the product cannot flow into the storage tank, a purge pump will start and mechanically fill the tank.

An automated computerized system will monitor and control the entire process, ensuring production of ethanol throughout a wide range of wind conditions. In the event of a power failure, a dump valve on the discharge of the compressor will automatically open, harmlessly discharging the gases to the atmosphere.

30 Wind Turbine

The wind turbine shall be the vertical axis type, mounted on the center mast. The buckets shall be fabricated from "Shearfil" and lined with aluminum in order to provide a radar signature. In the event of higher than expected winds creating an over speed condition of the turbine, a shear pin fails permitting the rotors to automatically fold.

The maximum speed at which the wind turbine will rotate is 23 RPM.

The maximum expected horsepower generated by the wind turbine is 1300 HP.

31 Bull Gear

A circular, spoked assembly (bull gear) is mounted to lower end of the wind turbine shaft and will rotate at the same RPM.

A roller chain, welded to the inner side of the bull gear, will engage sprockets mounted on the ends of the vertical drive shafts extending upward from the generator and process gas compressor.

The diameter of the bull gear and sprockets will permit a gear step-up ratio for the drive shafts of 18:1.

32 Electrolyser

The electrolyser is the device in which hydrogen and oxygen are separated from the surrounding seawater.

The center mast forms the outer casing of the electrolyser and consists of 6 inch diameter, schedule 40 pipe. The center mast is structurally fastened to the center tank through welded flange plates mounted top and bottom.

The center mast extends from below the bottom of the center flotation tank, up through the tank, to the top of the wind turbine. The upper section of the mast supports the upper bearing of the wind turbine. The bottom end of the center mast is open to the sea; the upper end is sealed. The total length of the mast is approximately 80 feet.

A two inch diameter pipe mounted through the middle of the 6 inch center mast shall form the inner core of the electrolyser. The bottom of the two inch pipe will be located approximately 20 feet above the lower end of the outer casing and is sealed with a pipe cap. The upper end is open and extends to within 12 inches of the top of the outer casing. A two inch section of pipe is welded to the lower end of the inner core and extends outward, through the wall of the outer casing and terminates in a flanged connection.

Positive and negative direct current electrodes will be mounted through the wall of the outer casing at such a level as to be immersed in the seawater during operation.

During normal operation of the system, the center mast will be under a vacuum permitting seawater to rise approximately 31 feet above the surface of the outside water.

33 Process Gas Compressor

The process gas compressor will be located in the inboard tank of the forward leg.

The inlet of the process gas compressor is connected to the electrolyser, causing a vacuum to be placed on the electrolyser when the compressor is operating.

The gas compressor will be mounted vertically, with the input shaft driven by a sprocket which engages the bull gear.

The gas compressor will be fabricated from stainless steel and will be non-sparking.

34 Electric Generator

The electric generator will be located in the inboard tank of the port leg.

The generator will be mounted vertically, with the input shaft driven by a sprocket which engages the bull gear.

The generator will be sized to produce 300 KW at 12 VDC. Output from the generator will power the electrolyser and the various process electrical systems.

35 Reactor

The reactor is a proprietary item of equipment and will be mounted on the port side of the forward pontoon.

The reactor is a pressure vessel in which the separated hydrogen and oxygen are combined with bottled CO_2 to form ethanol.

The reactor will be covered with thermal insulation and equipped with a relief valve to prevent over-pressurization.

36 Condenser

The condenser will be mounted on the starboard side of the forward pontoon and will be of the naturally air cooled type.

The condenser will receive gaseous ethanol being discharged from the reactor and condense it to liquid form.

The condenser will be equipped with a relief valve to prevent over-pressurization.

39 Process Control System

The entire process will be controlled by an automated computer system. The control and monitoring system will receive inputs from temperature, pressure, and flow instrumentation located throughout the process and make necessary control adjustments to ensure quality of the product produced. The system will be designed to permit continuous production of ethanol throughout a wide range of wind conditions.

The center of the control and monitoring system shall be a computer, mounted in the dome atop the center structural (electrolyser) mast.

Instrumentation will be rated for the system conditions in which it is installed.

38 Bladder Storage Tank

The ethanol product will be stored in a floating bladder style tank, tethered at the stern of the platform.

The bladder is sized to contain 40,000 gallons and is equipped with towing grommets in each corner. The bladder is a commercial, off the shelf, item; its material is nylon reinforced nitrile rubber.

4 PLATFORM EQUIPMENT

41 Cleats

Three cleats will be supplied. They will be mounted at the halfway point of each pontoon for the purpose of securing the anchor lines.

The cleats will be rated at a line pull of 1000 pounds.

42 Mooring Bitt

A mooring bitt will be provided at the end of the forward pontoon for towing purposes.

The mooring bitt will be rated at 2000 pounds line pull.

43 Bladder Tank Towing Bitts

Towing bitts for the bladder tank will be mounted on the platforms located at the ends of the port and starboard pontoons.

Each towing bitt will be rated at 1000 pounds line pull.

44 Electrical System

A 12 Volt, Direct Current electrical system will be installed to provide power for all platform installed systems including navigation lights, and communications/ telemetry.

45 Storage Batteries

12 Volt storage batteries will be provided as a power supply for all on-board electrical users.

The batteries will be kept charged by the electrical generator so as to be available for emergency back-up power for a period of 24 hours in the event of a failure.

46 Navigational Lighting

Lights shall be provided on six foot masts mounted on each flotation tank located at the ends of the pontoons and on the center mast.

Starboard pontoon will have a green light

Port pontoon will have a red light.

Forward pontoon and center mast will have white lights.

5 MAJOR EQUIPMENT LIST

Number in ( ) refers to paragraph.

51 Platform Structure

Tension Cables (21)

6x37 Class Fiber Core
Extra Improved Plow Steel
Right Regular lay
¾ in. Diameter
47,600 lbs Break Strength
Supplier – Web Rigging

Flotation Tanks (22)

Quantity – 7
7 ft. Diameter
30 ft.-6 in. Height Overall
18 ga. Corrugated Steel
Supplier – Pax Steel Products, Inc.

52 Process Equipment

Wind Turbine (30)

Quantity – 1
Vertical Axis Style
Material – Shearfil lined with Aluminum
Maximum RPM – 23
Supplier – B.E.A.R. Oceanics

Bull Gear (31)

Quantity – 1
Diameter 30 ft.
Supplier – B.E.A.R. Oceanics

Process gas Compressor (33)

Quantity – 1
Inlet Conditions
14.7 psia, 70^0F
10,000 SCFM Oxygen, Hydrogen, Oxygen
Outlet Conditions
150 psig, 300^0F
Stainless Steel Construction
Non-sparking
Vertical Mounting

Remote Driven, Sprocketed Drive Shaft
Supplier – Duraflow

Electric Generator (34)
Quantity – 1
Exterior Marine Service
300KW
12VDC
Remote Driven, Sprocketed Drive Shaft
Supplier – General Electric

Reactor (35) - **Proprietary**
Quantity – 1
Stainless Steel Construction
Inlet Conditions
150 PSIG, 300^0F
2000 ACFM Hydrogen, Oxygen
Outlet Conditions
150 PSIG, 500^0F
Flanged Connections

Condenser (36)
Quantity – 1
Inlet Conditions
150 psig, 500^0F
800 ACFM Gaseous Ethanol
Outlet Conditions
5 psig, 150^0F
30 GPM Ethanol
Stainless Steel Construction
Flanged Connections
Air Cooled
Supplier – B.E.A.R. Oceanics

53 Platform Equipment

Anchor Cleats (41)
Quantity – 3
Cast Steel Construction
Line Pull 1000 lbs.
Supplier – Anchor Marine & Industrial Supply or equal

Battery Charger (45)
Quantity – 1
12 Volt

Storage Batteries (46)
12 Volt
Type Sealed